



INTERNSHIP REPORT

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DEPARTMENT	Computer Science and Engineering
INTERNSHIP DURATION	15th June 2026 - 14th July 2026
ORGANISATION	Systech Solutions Pvt Ltd, Chennai
DOMAIN	Data Engineering
UNDER SUPERVISION OF	Swati Venkatesh , DineshKumar Sowrirajan

S.NO	TABLE OF CONTENTS	PG.NO
1.	Executive Summary	3
2.	Company Overview	3
3.	Internship Overview	4
4.	Weekly overview of Internship Activities	5
5.	Employee Master Data Management(MDM) Project	11
6.	Tools and Technologies Used	15
7.	Challenges faced	16
8.	Learning Outcomes	16
9.	Technical Skills Acquired	17
10.	Conclusion	17

Executive Summary

During my one-month internship at **Systech Solutions Pvt. Ltd.**, I gained practical exposure to **Data Engineering** through hands-on learning with **Snowflake** and **SQL**. The internship began with understanding the business domain by exploring the **Alliant Insurance** website, followed by learning Snowflake, SQL, and Streamlit. The major assignment was the **Employee Master Data Management (MDM) Project**, where I integrated employee data from multiple source systems to create trusted **Golden Records**. Through this project, I gained experience in database design, data loading, data integration, data standardization, and data quality validation, while also learning **Common Table Expressions (CTEs)** and the fundamentals of **Window Functions**. Overall, the internship strengthened my technical skills and provided valuable insight into real-world data engineering practices.

Company Overview

Systech Solutions Pvt. Ltd. is a technology consulting and digital transformation company specializing in **Data Engineering, Data Analytics, Cloud Computing, Artificial Intelligence (AI), Business Intelligence (BI), and Enterprise Application Development**. The company helps organizations build scalable data solutions by integrating, managing, and analyzing enterprise data to support informed business decisions. It provides services in cloud data warehousing, data integration, ETL, Master Data Management (MDM), and analytics across various industries. By leveraging modern technologies and industry best practices, Systech Solutions enables businesses to improve data quality, streamline operations, and accelerate digital transformation. The company promotes a collaborative and innovation-driven work environment, encouraging continuous learning and practical application of emerging technologies. During my internship, I had the opportunity to work with the Data Engineering team and gain

hands-on experience with industry-standard tools and real-world data engineering practices.



Internship Overview

The internship was undertaken at Systech Solutions Pvt. Ltd., Chennai, as a Data Engineering Intern from 15 June 2026 to 14 July 2026 under the guidance of Ms. Swathi, Mr. DineshKumar Sowrirajan. The primary objective of the internship was to gain practical exposure to Data Engineering concepts and understand how enterprise data is collected, integrated, processed, and managed using modern cloud technologies.

The internship followed a structured learning approach, beginning with business understanding and technology exploration before moving on to project implementation. During the initial phase, I explored the Alliant Insurance website to understand its business domain and documented my observations. This was followed by learning

fundamental concepts of Snowflake, SQL, and Streamlit, while strengthening my SQL skills through regular HackerRank practice.

The major assignment during the internship was the Employee Master Data Management (MDM) Project, where I worked on integrating employee data from multiple source systems using Snowflake and SQL. Throughout the project, I gained practical experience in database design, data loading, data standardization, and data quality validation. The internship provided valuable industry exposure and significantly enhanced my technical knowledge, analytical thinking, and problem-solving skills in the field of Data Engineering.

Weekly Overview of Internship Activities

Week 1 - Company Induction and Technology Exploration

- Introduction to Systech Solutions Pvt. Ltd. and the Data Engineering team.
- Overview of the company's services, projects, and work culture.
- Explored the Alliant Insurance website to understand its business domain and services.
- Prepared a document summarizing the business understanding of Alliant Insurance.
- Introduction to Snowflake and its cloud data warehousing architecture.
- Learned the fundamentals of relational databases and SQL.
- Explored the Snowflake user interface, databases, schemas, tables, and warehouses.
- Gained basic exposure to Streamlit and its role in building data applications.
- Practiced basic SQL queries to strengthen database concepts.

About > Why Alliant

Why Alliant?

- Alliant Insurance Services has the highest net inflow of industry-leading broker talent among the nation's leading insurance brokerages.



WEEK 2 - Employee Master Data Management (MDM) Project Initiation

- Introduction to the Employee Master Data Management (MDM) project.
- Understood the project requirements and business problem statement.
- Analyzed the employee dataset received from multiple source systems.
- Explored the structure and quality of the source data.
- Designed the database and created required objects in Snowflake.
- Created databases, schemas, tables, file formats, and stages.
- Loaded source data into Snowflake using appropriate data loading techniques.
- Practiced SQL queries and data transformation techniques.
- Strengthened SQL skills through regular HackerRank practice.



```

1 CREATE DATABASE EMPLOYEE_DB;
2 create schema DBO;
3
4
5 SELECT CURRENT_DATABASE(), CURRENT_SCHEMA();
6
7 CREATE OR REPLACE FILE FORMAT FF_EMPLOYEE_CSV
8 TYPE = CSV
9 FIELD_DELIMITER = ','
10 SKIP_HEADER = 1
11 FIELD_OPTIONALLY_ENCLOSED_BY = '"'
12 TRIM_SPACE = TRUE
13 EMPTY_FIELD_AS_NULL = TRUE
14 ✦ NULL_IF = ('NULL', '');
15
16 CREATE OR REPLACE STAGE STG_EMPLOYEE_FILES
17 FILE_FORMAT = FF_EMPLOYEE_CSV;
18
19 show stages;
20
21 LIST @STG_EMPLOYEE_FILES;

```

STAGE EMPLOYEE

Table Chart Pivot									
800 rows 43ms									
EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE	DEPARTMENT	CITY	SOURCE_SYSTEM		
E0001	Dee	Kumar	ajay.kumar159@company....	9125282...	Finance	Delhi	ACTIVEDIRECTOR...	25.0%	
E0004	Arun	Joseph	ajay.patel137@company.c...	9129163...	Marketing	Mumbai	CRM	25.0%	
+98 more	+28 more	+13 more	+98 more	+98 more	+4 more	+4 more	+2 more		
1	E0001	David	Verma	david.verma1@company.com	9637637234	Finance	Chennai	HRM	
2	E0001	David	Verma	david.verma1@company.com	9637637234	Finance	Chennai	ACTIVEDIRECTORY	
3	E0001	David	Verma	david.verma1@company.com	null	null	Chennai	CRM	
4	E0001	David	Verma	david.verma1@company.com	9637637234	null	Chennai	PAYROLL	
5	E0002	Arun	Verma	arun.verma2@company.com	9338898955	Marketing	null	HRM	
6	E0002	Arun	Verma	arun.verma2@company.com	9338898955	Marketing	Hyderabad		

DATASET

#	A	B	C	D	E	F	G
1	USER_ID	FIRST_NAME	LAST_NAME	EMAIL	LOCATION		
2	E0220	Sneha	Shah	sneha.shah220@company.com	Chennai		
3	E0175	Nisha	Smith	nisha.smith175@company.com	Mumbai		
4	E0164	Meena	Shah	meena.shah164@company.com	Delhi		
5	E0198	Mee	Gupta	meena.gupta198@company.com	Delhi		
6	E0137	Ajay	Patel	ajay.patel137@company.com	Delhi		
7	E0193	Sneha	Das	sneha.das193@company.com	Pune		
8	E0202	Ajay	Roy	ajay.roy202@company.com	Pune		
9	E0119	Arun	Das	arun.das119@company.com	Chennai		
10	E0008	Joh	Das	john.das8@company.com	Mumbai		
11	E0197	Sneha	Roy	sneha.roy197@company.com	Mumbai		
12	E0095	David	Iyer	david.iyer95@company.com	Bengaluru		
13	E0078	Meena	Gupta	meena.gupta78@company.com	Chennai		
14	E0200	Sneha	Joseph	sneha.joseph200@company.com	Chennai		
15	E0018	Arun	Lee	arun.lee18@company.com	Delhi		
16	E0231	Rahul	Rao	rahul.rao231@company.com	Pune		
17	E0088	Priya	Lee	priya.lee88@company.com	Delhi		
18	E0241	Nis	Kumar	nisha.kumar241@company.com	Pune		
19	E0234	Rahul	Gupta	rahul.gupta234@company.com	Mumbai		
20	E0184	Nisha	Patel	nisha.patel184@company.com	Hyderabad		
21	E0153	Meena	Shah	meena.shah153@company.com	Pune		
22	E0133	Rahul	Iyer	rahul.iyer133@company.com	Hyderabad		
23	E0026	Deepa	Iyer	deepa.iyer26@corp.com	Pune		
24	E0189	Anita	Singh	anita.singh189@company.com	Mumbai		
25	E0116	Ani	Verma	anita.verma116@company.com	Bengaluru		
26	E0154	Deepa	Roy	deepa.roy154@company.com	Delhi		
27	F0192	Deena	Iyer	deena.iyer192@company.com	Delhi		

WEEK 3 - Data Integration and Golden Record Creation

- Integrated employee data from multiple source systems.
- Performed data cleansing and standardization to improve data quality.
- Applied matching rules to identify duplicate employee records.
- Implemented survivorship rules to determine the most reliable data values.
- Created Golden Records representing a single trusted version of each employee.
- Learned and implemented Common Table Expressions (CTEs).
- Learned the fundamentals of Window Functions for data processing.
- Generated a Data Quality Report to validate the processed data.
- Successfully completed the Employee Master Data Management (MDM) project.

MDM

Workspaces

Databases

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MDMnew.sql

STAGE_EMPLOYEE.sql

TGT_EMPLOYEE.sql

Untitled 1.sql

HRM.sql

DataQuality.sql

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TablesM1.sql

StagetotableM1.sql

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WITH cleaned AS (
    SELECT
        EMPLOYEE_ID,
        NULLIF(TRIM(FIRST_NAME), '') AS FIRST_NAME,
        NULLIF(TRIM(LAST_NAME), '') AS LAST_NAME,
        NULLIF(TRIM(EMAIL), '') AS EMAIL,
        NULLIF(TRIM(PHONE), '') AS PHONE,
        NULLIF(TRIM(DEPARTMENT), '') AS DEPARTMENT,
        NULLIF(TRIM(CITY), '') AS CITY,
        CASE SOURCE_SYSTEM
            WHEN 'HRMS' THEN 1
            WHEN 'PAYROLL' THEN 2
            WHEN 'CRM' THEN 3
            ELSE 4
        END AS SRC_PRIORITY
    FROM STAGE_EMPLOYEE
),

| Ctrl+I to generate
fn AS ( -- FIRST_NAME
    SELECT EMPLOYEE_ID, FIRST_NAME,
        ROW_NUMBER() OVER (PARTITION BY EMPLOYEE_ID ORDER BY CNT DESC, MIN_PRI ASC) AS RN
    FROM (
        SELECT EMPLOYEE_ID, FIRST_NAME, COUNT(*) AS CNT, MIN(SRC_PRIORITY) AS MIN_PRI
        FROM cleaned
        WHERE FIRST_NAME IS NOT NULL
        GROUP BY EMPLOYEE_ID, FIRST_NAME
    ) t
),

ln AS ( -- LAST NAME
    SELECT EMPLOYEE_ID, LAST_NAME,
        ROW_NUMBER() OVER (PARTITION BY EMPLOYEE_ID ORDER BY CNT DESC, MIN_PRI ASC) AS RN
    FROM (
        SELECT EMPLOYEE_ID, LAST_NAME, COUNT(*) AS CNT, MIN(SRC_PRIORITY) AS MIN_PRI

```

MASTER RECORD

```
SELECT
    i.EMPLOYEE_ID,
    fn.FIRST_NAME,
    ln.LAST_NAME,
    fn.FIRST_NAME || ' ' || ln.LAST_NAME AS FULL_NAME,
    em.EMAIL,
    ph.PHONE,
    dp.DEPARTMENT,
    ct.CITY
FROM ids i
LEFT JOIN fn ON fn.EMPLOYEE_ID = i.EMPLOYEE_ID AND fn.RN = 1
LEFT JOIN ln ON ln.EMPLOYEE_ID = i.EMPLOYEE_ID AND ln.RN = 1
LEFT JOIN em ON em.EMPLOYEE_ID = i.EMPLOYEE_ID AND em.RN = 1
LEFT JOIN ph ON ph.EMPLOYEE_ID = i.EMPLOYEE_ID AND ph.RN = 1
LEFT JOIN dp ON dp.EMPLOYEE_ID = i.EMPLOYEE_ID AND dp.RN = 1
LEFT JOIN ct ON ct.EMPLOYEE_ID = i.EMPLOYEE_ID AND ct.RN = 1
ORDER BY i.EMPLOYEE_ID;
```

WEEK 4 - Project Review and Skill Enhancement

- Reviewed the completed MDM project and verified the project outputs.
- Refined SQL queries and validated the generated Golden Records.
- Continued practicing SQL through HackerRank challenges.
- Strengthened understanding of SQL joins, aggregate functions, CTEs, and Window Functions.
- Documented the project implementation and key learnings.
- Reviewed data engineering concepts learned throughout the internship.
- Improved analytical thinking and problem-solving skills through continuous practice.
- Gained confidence in applying Snowflake and SQL to real-world data engineering scenarios.

- Successfully completed the internship with enhanced technical and practical knowledge.

	# TOTAL_RECORDS_RECEIVED	# DUPLICATE_RECORDS_IDENTIFIED	# GOLDEN_RECORDS_CREATED	# REJECTED_OR_INVALID_RECORDS
1	800	550	239	200

Employee Master Data Management (MDM) Project

The Employee Master Data Management (MDM) Project was assigned to provide practical exposure to handling enterprise data from multiple source systems. The objective of the project was to consolidate employee information from different systems into a single, accurate, and trusted master record. Since employee data often exists in multiple applications with varying formats and inconsistencies, the project focused on improving data quality through cleansing, standardization, matching, and consolidation techniques. The implementation was carried out using Snowflake and SQL, enabling a practical understanding of cloud-based data engineering workflows and Master Data Management concepts.

Problem Statement:

Title: Employee Master Data Management (MDM) Exercise

Objective

The organization receives employee information from multiple source systems. Due to differences in data formats and quality, the same employee may appear multiple times with inconsistent information.

Your task is to consolidate employee data, create a single Golden Record for each employee, and load the cleansed data into the target employee master table.

Source Systems

You are provided with employee data from the following source systems:

- HRMS
- Payroll
- Active Directory
- CRM

Tasks

1. Load all four source datasets into a staging table named:
 - STAGE_EMPLOYEE
2. Perform MDM activities, including:
 - Standardize employee names.

Database Design

The first step in the implementation of the Employee Master Data Management (MDM) project was designing the database structure in **Snowflake**. A dedicated database and schema were created to organize the project objects efficiently. Separate tables were created for each source system (**HRMS, Payroll, Active Directory, and CRM**) to store the raw employee data. A staging table (**STAGE_EMPLOYEE**) was then created to consolidate records from all source systems into a common structure. Finally, a target table (**TGT_EMPLOYEE**) was designed to store the cleansed and consolidated Golden Records.

CREATE DATABASE EMPLOYEE_DB;

CREATE SCHEMA DBO;

Results (1 hour ago)

Table	Chart	Pivot	4 rows	35ms			
id	name	size	md5	last_modified			
1	stg_employee_files/ActiveDirectory.csv	10192	72ca113e3c6a60ed131012d02b908ed3	Wed, 1 Jul 2026 11:08:49 GMT			
2	stg_employee_files/CRM.csv	12336	44bc7efd0aa5a13609f736fb74191563	Wed, 1 Jul 2026 11:08:49 GMT			
3	stg_employee_files/HRM.csv	13760	c74cdd883ecf6250d58ec823d999336b	Wed, 1 Jul 2026 11:08:49 GMT			
4	stg_employee_files/payroll.csv	12256	4e13945c67d6f057ff1353212b59b95c	Wed, 1 Jul 2026 11:08:49 GMT			

Data Loading and Source Integration

The employee data was provided in **Microsoft Excel** format and contained information from four different source systems. Each dataset was imported into Snowflake using file formats, stages, and the **COPY INTO** command. After loading the data into individual source tables, the records were integrated into the **STAGE_EMPLOYEE** table using

SQL queries. This created a unified dataset that served as the foundation for further data cleansing and transformation.

COPY INTO HRMS

FROM @STG_EMPLOYEE_FILES;

Data Cleansing and Standardization

Since the employee information originated from multiple systems, the data contained inconsistencies such as missing values, extra spaces, and different naming conventions. Data cleansing was performed to remove unnecessary spaces, standardize names, validate email addresses and phone numbers, and handle NULL values. These transformations improved data consistency and ensured that all records followed a common format before duplicate identification.

NULLIF(TRIM(FIRST_NAME),")

AFTER CLEANSING

Table Chart Pivot												
239 rows 229ms												
	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	FULL_NAME	EMAIL	PHONE	LOCATION					
	E0001	Arun	Kumar	Arun Lee	ajay.iyer41@corp.com	91100...	Finan...	0.4%	9.2%	9.6%	1.7%	0.4%
	E0002	Rahul	Joseph	Arun V...	ajay.joseph10@compa...	91181...	Mark...	0.4%	9.2%	8.8%	1.7%	0.4%
	+98 more	+24 more	+13 more	+98 more	+98 more	+98 more	+4 more					
1	E0001	David	Verma	David Verma	david.verma1@company.com	9637637234	Finance					
2	E0002	Arun	Verma	Arun Verma	arun.verma2@company.com	9338898955	Marketing					
3	E0003	Ravi	Das	Ravi Das	ravi.das3@company.com	9962388353	Finance					
4	E0004	Priya	Roy	Priya Roy	priya.roy4@company.com	9184748096	Finance					
5	E0005	Deepa	Nair	Deepa Nair	deepa.nair5@company.com	9294138474	Operations					
6	E0006	Meena	Miller	Meena Miller	meena.miller6@company.com	9284339916	Finance					
7	E0007	John	Verma	John Verma	john.verma7@company.com	9437977780	Finance					
8	E0008	John	Das	John Das	john.das8@company.com	9299675031	Finance					
9	E0009	Nis	Shah	Nis Shah	nisha.shah9@company.com	9359201416	Operations					
10	E0010	Ajay	Joseph	Ajay Joseph	ajay.joseph10@company.com	9947783944	HR					
11	E0011	Ravi	Singh	Ravi Singh	ravi.singh11@company.com	9793073200	IT					
12	E0012	Ravi	Patel	Ravi Patel	ravi.patel12@corp.com	9761523329	Marketing					

Matching Rules

To identify duplicate employee records, predefined matching rules were applied based on common employee attributes. Records were compared using **Employee ID**, **Email Address**, and **Employee Name**. These rules ensured that records belonging to the same employee were grouped together, even if they originated from different source systems.

Survivorship Rules

After identifying duplicate records, survivorship rules were implemented to determine the most reliable value for each employee attribute. Source systems were assigned priorities based on data reliability, with **HRMS** considered the most trusted source, followed by **Payroll**, **CRM**, and **Active Directory**. These rules ensured that the highest-quality information was selected while creating the final master record.

Golden Record Creation

Once the matching and survivorship rules were applied, duplicate employee records were consolidated into a single trusted record known as the **Golden Record**. Advanced SQL techniques, including **Common Table Expressions (CTEs)** and **Window Functions**, were used to process the data efficiently. The final Golden Records were stored in the **TGT_EMPLOYEE** table, providing a single, accurate version of each employee's information.

Data Quality Validation

After generating the Golden Records, a Data Quality Report was prepared to evaluate the overall quality of the processed data. Validation checks were performed to identify duplicate records, invalid values, and the total number of Golden Records created. This

step ensured that the final dataset met the required quality standards and accurately represented the employee master data.

Results (3 hours ago)				
Table Chart Pivot				
	# TOTAL_RECORDS_RECEIVED	# DUPLICATE_RECORDS_IDENTIFIED	# GOLDEN_RECORDS_CREATED	# REJECTED_OR_INVALID_RECORDS
1	800	550	239	200

Project Outcome

The Employee Master Data Management (MDM) project was successfully implemented using **Snowflake** and **SQL**. Employee information from multiple source systems was integrated, cleansed, standardized, and consolidated into a single trusted master dataset. The implementation improved data consistency, reduced duplication, and ensured reliable employee information through Golden Record creation and data quality validation. This project provided valuable hands-on experience in database design, data integration, SQL development, and cloud-based data engineering practices.

Tools and Technologies Used

TOOLS/TECHNOLOGY	PURPOSE
Snowflake	Used as the cloud data warehouse to create databases, tables, load source data, and implement the MDM solution.
SQL	Used for data querying, transformation, integration, and implementation of matching and survivorship rules.
MS Excel	Used as the source of employee datasets received from multiple systems.
Streamlit	Explored to understand the basics of developing data-driven web applications.
HackerRank	Used to practice SQL queries and improve problem-solving and query-writing skills.

Challenges Faced

During the internship, I encountered several challenges while learning new technologies and implementing the Employee Master Data Management (MDM) project. Since Snowflake was new to me, understanding its architecture, database objects, and data loading process required continuous learning and practice. Integrating employee data from multiple source systems was another challenge due to differences in data formats, missing values, and duplicate records. I also learned advanced SQL concepts such as Common Table Expressions (CTEs) and Window Functions, and applied them to implement data cleansing, matching, survivorship rules, and Golden Record creation. Overcoming these challenges improved my analytical thinking, problem-solving skills, and confidence in working with modern data engineering tools and techniques.

Learning Outcomes

The internship provided me with valuable practical exposure to Data Engineering and helped me develop both technical and analytical skills. Through hands-on learning and project implementation, I gained a better understanding of industry practices and modern data engineering workflows.

The key learning outcomes include:

- Developed a strong foundation in Data Engineering and SQL.
- Gained hands-on experience with Snowflake for cloud data warehousing.
- Learned data loading, integration, and transformation techniques.
- Understood Master Data Management (MDM) concepts, including matching and survivorship rules.
- Learned to use CTEs and the fundamentals of Window Functions in SQL.
- Improved problem-solving, analytical thinking, and data quality validation skills.

Overall, the internship enhanced my confidence in applying Data Engineering concepts and prepared me for future industry projects.

Technical Skills Acquired

During my internship, I gained practical experience with several tools and technologies used in Data Engineering. The key technical skills acquired include:

- **SQL** – Data querying, transformation, and analysis.
- **Snowflake** – Database creation, data loading, and cloud data warehousing.
- **Master Data Management (MDM)** – Data cleansing, matching, survivorship rules, and Golden Record creation.
- **Database Design** – Designing staging and target tables.
- **Data Integration** – Consolidating data from multiple source systems.
- **CTEs and Window Functions** – Implementing advanced SQL techniques for data processing.

Conclusion

The internship at Systech Solutions Pvt. Ltd. provided me with valuable practical exposure to Data Engineering and modern data management practices. Throughout the internship, I gained hands-on experience with Snowflake, SQL, and Master Data Management (MDM) while working on the Employee Master Data Management project. This experience enhanced my technical knowledge, analytical thinking, and problem-solving skills. Overall, the internship helped me bridge the gap between academic learning and industry practices, and it has strengthened my confidence in pursuing a career in the field of Data Engineering.